

Common requirements for the assignments

You need to extend the program and the documentation of the 2nd assignment with the extensions provided in this document. The program is only acceptable if you also have the documentation.

Similarly to the 2nd assignment, there may be some things that are not defined. In these cases you are free to design and implement them as you see fit.

The documentation should include these additions

- an extended class diagram
- a state diagram
- two sequence diagrams

The code should include

- javadoc comments for each class and method (except getters and setters)
- unit tests for each of the important methods (not getters or setters)

7.1 Additional functionality

Psychological tensions and loyalties are introduced into the Borderland to create room for player agency and strategic play.

Every character has a behaviour type – Rational, Aggressive, Cautious, or Unstable – that determines the manner in which he or she responds to fear and game type. Each influences them behind the scenes to influence their decision-making, fear gain, and play. For example, Cautious characters avoid dangerous games, and Aggressive characters ignore fear but utilize stamina.

The system also has trust-based relations that depreciate over time without being solidified by mutual success or card trades. Betrayals invoke fear and alter the behaviour of the characters.

7.2 State diagram

Design a UML state machine that starts every character in the Rational state and allows transitions to Aggressive, Cautious, or Unstable based on simple triggers: fear spikes push Rational to Cautious and Cautious to Unstable, betrayal drives Rational/Cautious to Aggressive, prolonged aggression with low stamina turns Aggressive to Unstable, while rest or trust-building can move Aggressive/Cautious back toward Rational and Unstable to Cautious to Rational.

7.3 Scenarios to model with sequence diagrams

7.3.1 Scenario 1: Typical game

Draw a sequence diagram for the “Start Game” interaction: the Borderland Manager instructs the Game Master to begin the next round; the Game Master randomly selects a Game from the Arena’s queue, notifies the Arena to set itself up, and then broadcasts a “game start” message to all Players, who each respond by checking their relevant stat (mental or physical) against the Game’s difficulty and returning either “win” or “lose”; the Game Master tallies results, updates each Player’s fear level, and awards cards to winners before reporting back to the Borderland Manager.

7.3.2 Scenario 2: Victory!

Create a sequence diagram depicting the moment a Player wins their last needed card: the Game object notifies the Game Master of the Player’s vic-

tory, the Game Master updates the Player’s card collection, discovers the set now contains all 52 unique cards, and immediately informs the Borderland Manager. The Manager validates the full deck, broadcasts an “exit granted” message, orders the Arena to open an exit path, and logs the event for spectators while the victorious Player sends a final “leave borderland” confirmation back to both the Game Master and Manager.









